MSG No. 112.259 $P\bar{4}2c$ [Type I, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,6]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,7,8]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,7,8]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,2]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[3,4]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, -x,\, \frac{3}{4}\right]$	[5,7]
4	$\left[0,\, x,\, \frac{3}{4}\right]$	[6,8]

Table 8: Wyckoff site: 4h, site symmetry: .2.

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, \frac{1}{4}\right]$	[1,3]
2	$\left[\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	[2,4]
3	$\left[y,\, \frac{1}{2},\, \frac{3}{4}\right]$	[5,8]
4	$\left[-y,\, \frac{1}{2},\, \frac{3}{4}\right]$	[6,7]

Table 9: Wyckoff site: 4i, site symmetry: .2.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{4}\right]$	[1,2]
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{4}\right]$	[3,4]
3	$\left[\frac{1}{2},\, -x,\, \frac{3}{4}\right]$	[5,7]
4	$\left[\frac{1}{2},\, x,\, \frac{3}{4}\right]$	[6,8]

Table 10: Wyckoff site: 4j, site symmetry: .2.

No.	position	mapping
1	$\left[0,\, y,\, \frac{1}{4}\right]$	[1,3]
2	$\left[0,\, -y,\, \frac{1}{4}\right]$	[2,4]
3	$\left[y,\, 0,\, \frac{3}{4}\right]$	[5,8]
4	$\left[-y,\, 0,\, \frac{3}{4}\right]$	[6,7]

Table 11: Wyckoff site: 4k, site symmetry: 2..

No.	position	mapping
1	$[0,\, 0,\, z]$	[1,4]
2	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	[2,3]
3	$[0,\, 0,\, -z]$	[5,6]
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	[7,8]

Table 12: Wyckoff site: 4l, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	[1,4]
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2} - z\right]$	[2,3]
3	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	[5,6]

continued ...

Table 12

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 13: Wyckoff site: `4m`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[7,8]`

Table 14: Wyckoff site: `8n`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[2]`
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[y,\ -x,\ -z]$	`[5]`
6	$[-y,\ x,\ -z]$	`[6]`
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[8]`